

Relationship Between Space Charge Behaviors and Trap Level Distribution in Polypropylene/Propylene Ethylene Rubber/ZnO Nanocomposites

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Abstract- This paper intends to reveal the mechanism of space charge suppression mechanism in polypropylene (PP)/propylene ethylene rubber (PER)/ZnO ternary nanocomposites. Space charge distribution and trap level density were measured by pulsed electro-acoustic (PEA) and modified thermally stimulated depolarization current (TSDC), respectively. PP/PER/ZnO nanocomposite with 0.5 phr ZnO nanoparticles exhibits improved space charge distribution. The addition of ZnO nanofillers contributes to the dissipation of hetero-charges, while increases the homo-charges. The reduced deep traps in the PP/PER/ZnO nanocomposites may contribute to the dissipation of the hetero-charges, thus significantly suppressing the formation of hetero-charges.

I. INTRODUCTION

Extruded cable for high voltage direct current (HVDC) power transmission has great value in a wide application range, such as the integration of renewable energies from offshore wind farms, connections between asynchronous power networks, mainland power supply to islands and offshore oil and gas platforms, city center with limited space and oversea power grid interconnections[1]. Extruded cables with cross-linked polyethylene (XLPE) insulation have been successfully applied in high-voltage direct current transmission in power systems because of its favorable combination of superior mechanical, thermomechanical and electrical properties. However, XLPE is difficult to be recycled when the cable reaches the end of its life because of the thermosetting properties of XLPE. Recent studies demonstrated that polypropylene (PP)/propylene ethylene rubber (PER) is suitable for the recyclable electrical insulation of extruded cables [2, 3].

The accumulation of space charge in polymer insulation under high DC electrical field is a formidable problem bringing an insulation to failure [4, 5]. The accumulation of space charge causes distortion of the local electric field and thus, partial discharge and subsequent insulation failure. Recently, many researchers have reported that the incorporation of nanoparticles can strongly improve the space charge distribution characteristics [6-8]. Furthermore, even though the space charge suppression mechanism and models based on shallow and deep traps have been proposed [9-16], it is not fully understood that which kind of traps play a leading role in suppressing space charge.

In this study, mechanism of space charge suppression is revealed by analyzing space charge characteristics and trap level distribution of PP/PER nanocomposites,

II. EXPERIMENTAL

A. Materials

Polypropylene T30s (Maoming Petrochemical Company) with a melt flow index of 3 g/10 min, density of 0.9 g/cm³ was used as the base material. PER VISTAMAXX 6102 (Exxon Mobil Chemical Company) was used to toughen PP. ZnO nanoparticles with an average diameter of about 50 nm were supplied by Aladdin.

B. Sample preparation

PP/PER/ZnO ternary composites were prepared by melt-blending in a Hapro mixer. The cavity temperatures were set at 453 K and blended for 30 min with a rotor speed of 60 r/min. The concentration of the PER in all the PP/PER/ZnO blends was 40 wt% and the ZnO concentration was 0, 0.5, 1, 3 and 5 phr, respectively. The detailed compositions are listed in Table I.

TABLE I PP/PER NANOCOMPOSITES WITH DIFFERENT ZnO NANOFILLER CONTENTS.

Abbreviation	Composition (weight ratio)		
	PP	PER	ZnO
0phr	60	40	0
0.5phr	60	40	0.5
1phr	60	40	1
3phr	60	40	3
5phr	60	40	5

The films were prepared by compression molding. Firstly the material was pre-melted using a curing press at 200 °C for 10 min; then the material were pressed at a pressure of ~20 MPa for 10 min, at last the material were pressed at 25 °C with a pressure of ~5 MPa for a further 10 min. The obtained film samples with thickness of ~300 μm were used to measure the space charge characteristics, trap level distribution, and crystalline structure of the samples.

C. Materials characterization

Space charge measurements were carried out using a pulsed electro-acoustic (PEA) system equipped with a Lecroy WaveRunner 610Zi (1 GHz, 20 GS/s digital oscilloscope). A

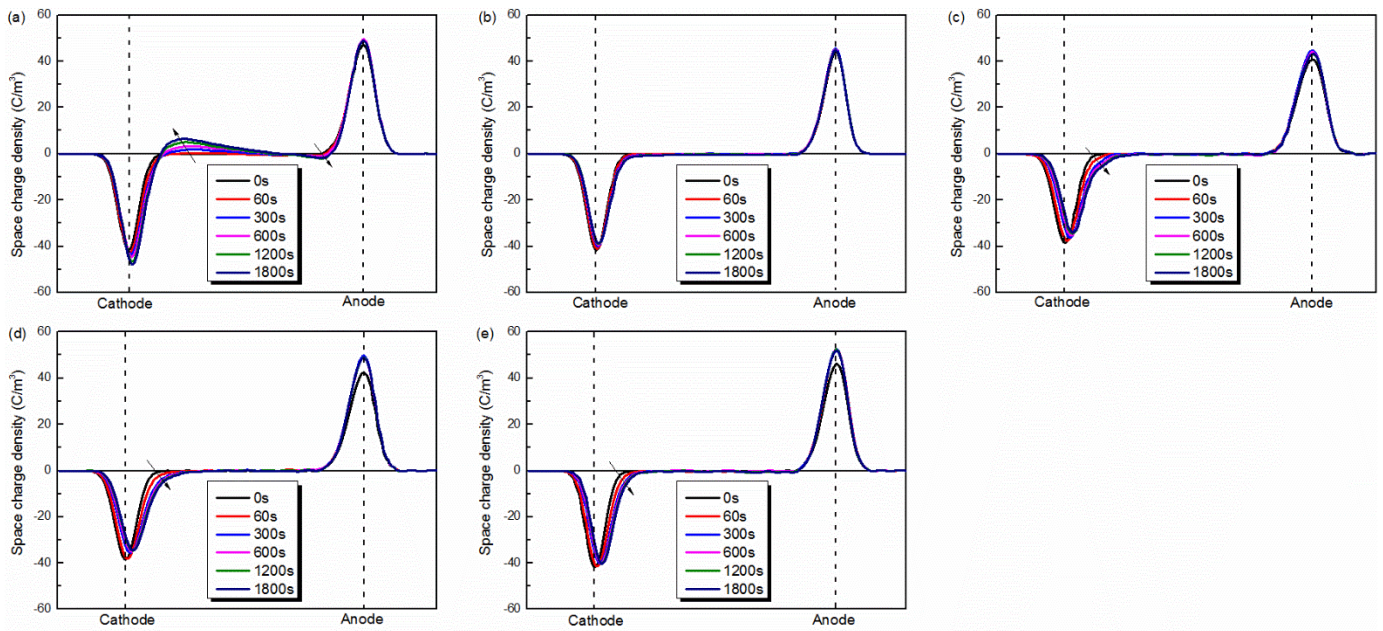


Fig. 1. Space charge distribution during polarization in (a) PP/PER, (b) 0.5 phr (c) 1 phr, (d) 3 phr, (e) 5 phr, subjected to a DC electric field of 60 kV/mm for 1800 s.

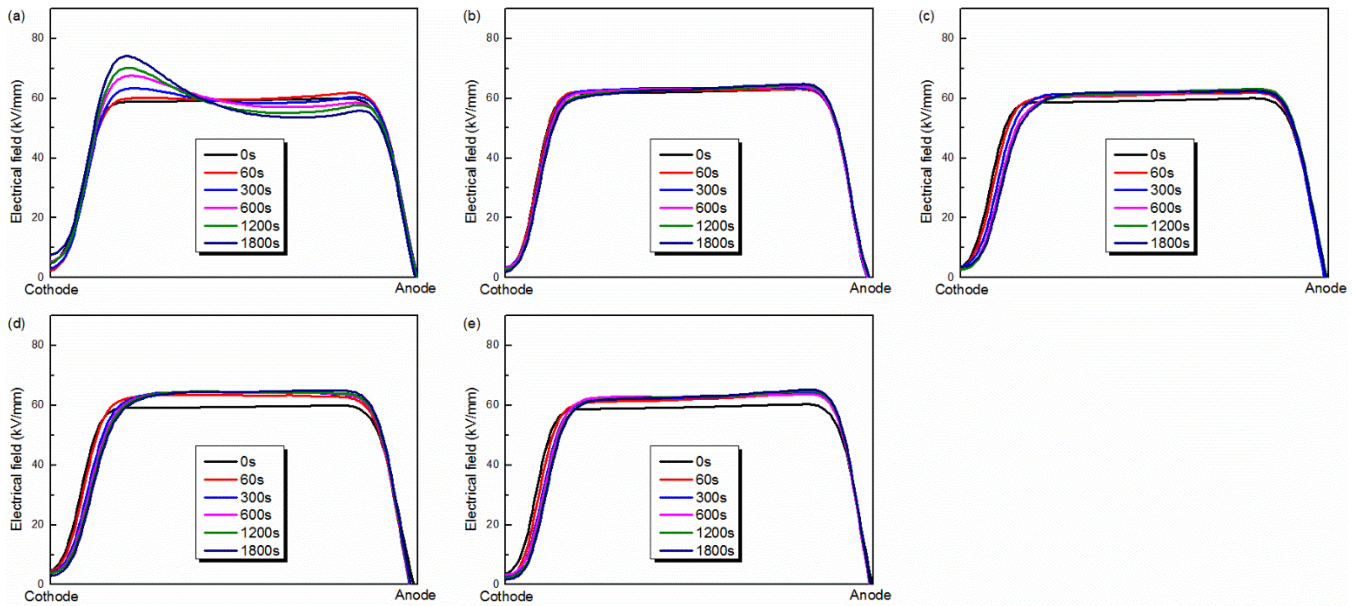


Fig.2. Electric field distribution during polarization in (a) PP/PER, (b) 0.5 phr (c) 1 phr, (d) 3 phr, (e) 5 phr, subjected to a DC electric field of 60 kV/mm for 1800 s.

DC electric field of 60 kV/mm, and an electric pulse with an amplitude of 800 V and duration of 7 ns were applied to the film samples for 1800 s. A piezoelectric transducer composed of 9 μm thick polyvinylidene fluoride film was used to change the detected acoustic signal into an electric signal. Meanwhile, calibration of acoustic wave attenuation was finished to obtain the accurate information about the space charge profiles [17].

The trap level profiles in PP and PP blends were calculated by the modified thermally stimulated depolarization current (TSDC) [18]. Gold electrodes with a diameter of 20 mm were sputtered on both sides of the samples for TSDC measurements. The sample was firstly polarized at 20 $^{\circ}\text{C}$ with a positive DC electric field 3 kV/mm for 20 min, and then cooled to -100 $^{\circ}\text{C}$ at a rate of 10 $^{\circ}\text{C}/\text{min}$. Then the poling voltage was removed and the sample was shorted for 10 min. The depolarization

current was measured from -100 to 100 °C at a heating rate of 3 °C/min.

A D8 advance diffract meter (Rigaku, Tokyo, Japan) equipped with a $\text{CuK}\alpha$ tube was used to investigate the crystalline structure of the samples over a range of diffraction angles $2\theta = 10\text{-}25^\circ$ with a scanning rate of 3 °/min at room temperature.

III. RESULTS AND DISCUSSION

A. Space charge characteristics

Figure 1 shows space charge distribution during polarization in (a) PP/PER, (b) 0.5phr (c) 1phr, (d) 3phr, (e) 5phr, subjected to a DC electric field of 60 kV/mm for 1800 s. The distance between the two vertical dotted lines is the thickness of the sample, which is equal to the distance from the cathode to the anode. Figure 1a illustrates that large number of hetero-charges accumulated around cathode in the neat PP/PER, while Figure 1b shows that the space charge distribution in 0.5phr PP/PER/ZnO was improved. There are almost no space charges in 0.5phr PP/PER/ZnO. In addition, with the increase of ZnO nanofiller content, the hetero-charges disappear while the homo-charges gradually increase.

The electric field distribution can be calculated from the space charge distribution by using Poisson's equation. Figure 2 shows that the distortion of electric field in the bulk of PP/PER/ZnO nanocomposites with different ZnO nanofiller contents due to the space charges formation in the materials. Figure 2a demonstrates that the field enhancement in the pure PP/PER is 14 kV/mm, which is about 23.3% of the average applied field ($E_{av}=60$ kV/mm) after 1800 s. Additionally, as shown in Figure 2b, the maximum electric field in 0.5phr PP/PER/ZnO is 4.7 kV/mm, which is only 7.8% of the average applied field. Such a uniform electric field is ideal for HVDC insulation application.

B. Trap level distribution

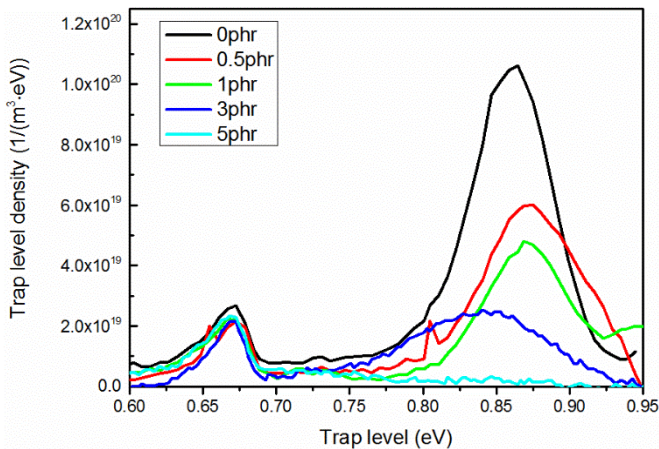


Fig. 3. Trap level distribution of PP/PER/ZnO nanocomposites with different ZnO nanofiller contents

Trap level distribution of PP/PER/ZnO nanocomposites with different ZnO nanofiller contents is presented in Figure 3. It indicates that all PP/PER/ZnO nanocomposites with different

ZnO nanofiller contents have two trap level peaks. The trap level peak located at about 0.67 eV is related to the motion of molecular chain of PER which is almost the same in all PP/PER/ZnO nanocomposites. In addition, with the increase of the ZnO nanofiller content, the trap level peak of deep traps obviously reduces.

TABLE II TRAP LEVEL DISTRIBUTION OF PP/PER NANOCOMPOSITES WITH DIFFERENT ZnO NANOFILLER CONTENTS.

Abbreviation	Trap level (eV)	Trap level density (1/m³·eV)
0phr	0.67	2.68×10^{19}
	0.86	1.06×10^{20}
0.5phr	0.67	2.16×10^{19}
	0.87	6.02×10^{19}
1phr	0.67	2.20×10^{19}
	0.87	4.81×10^{19}
3phr	0.67	2.54×10^{19}
	0.84	4.81×10^{19}
5phr	0.67	2.34×10^{19}
	/	/

C. Crystal structure analysis

In the case of isotactic PP, there are three well-known crystal structures, monoclinic (α), trigonal (β), and triclinic (γ), which forms depending on the melting history, crystallization temperature, pressure and cooling rate as well as the presence of nanoparticles. Figure 4 shows the XRD patterns of PP/PER/ZnO nanocomposites with different ZnO nanofiller contents. PP/PER has five main crystalline diffraction peaks at 2θ values of 14.0, 16.8, 18.5, 21.7, and 21.8 °, corresponding to the 110, 040, 130, 131, and 111 lattice planes, respectively, which indicates that the addition of PER did not influence the crystal structure of PP. In addition, the doping of nano-ZnO into PP/PER induces the formation of β phase PP (at $2\theta = 16^\circ$ shown by the arrows in Figure 4).

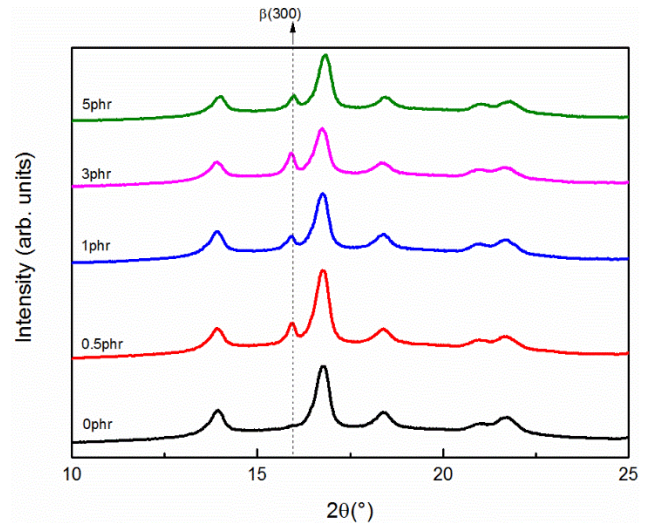


Fig. 4. XRD spectra of PP/PER/ZnO nanocomposites with different ZnO nanofiller contents

D. Space charge suppression mechanism

Space charge distribution is the direct result of charge trapping/detrapping and transport behavior in polymer dielectrics. The reduced deep traps in the PP/PER/ZnO nanocomposites may contribute to the dissipation of the hetero-charges, thus significantly suppressing the formation of hetero-charges. However, electrons around the cathode will inject into materials easily due to the fact that deep trap density decrease with increasing content of ZnO nanoparticle.

IV. CONCLUSION

In conclusion, by adopting a ZnO nanoparticle, we have fabricated a PP/PER/ZnO nanocomposites. The addition of ZnO nanofiller contribute to the dissipation of hetero-charges, while increase the home-charges. The reduced deep traps in the PP/PEC/ZnO nanocomposites lower potential barrier for charge injection, and contribute to electrons injected from the cathode. Furthermore, the reduced deep traps would promote the dissipation of hetero-charges, thus suppressing the formation of space charge to a significant degree.

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